

# LangChain Complete Cheat Sheet (FAANG-Level)

## 1. PROMPTS MODULE (Old)

```
from langchain.prompts import (
    PromptTemplate, ChatPromptTemplate, FewShotPromptTemplate,
    FewShotChatMessagePromptTemplate, MessagesPlaceholder,
    SystemMessagePromptTemplate, HumanMessagePromptTemplate,
    AIMessagePromptTemplate)
```

## 1. PROMPTS MODULE (New)

```
from langchain_core.prompts import (
    PromptTemplate, ChatPromptTemplate, MessagesPlaceholder)
```

## 2. LLMs & CHAT MODELS

```
from langchain_openai import ChatOpenAI, OpenAI
from langchain_community.llms import HuggingFaceHub
from langchain_community.chat_models import ChatOllama
```

## 3. RUNNABLES

```
from langchain_core.runnables import (
    RunnableSequence, RunnableLambda,
    RunnableParallel, RunnablePassthrough)
```

## Old Chains

```
from langchain.chains import (
    LLMChain, SequentialChain, SimpleSequentialChain)
```

## 4. MEMORY

```
from langchain.memory import (
    ConversationBufferMemory,
    ConversationSummaryMemory,
    ConversationBufferWindowMemory)
```

## 5. DOCUMENT LOADERS

```
from langchain_community.document_loaders import (
    PyPDFLoader, TextLoader, WebBaseLoader,
    DirectoryLoader, CSVLoader)
```

## 6. TEXT SPLITTERS

```
from langchain.text_splitter import (
    RecursiveCharacterTextSplitter,
    CharacterTextSplitter)
```

## 7. EMBEDDINGS

```
from langchain_openai import OpenAIEmbeddings
from langchain_community.embeddings import HuggingFaceEmbeddings
```

## 8. VECTOR STORES

```
from langchain_community.vectorstores import FAISS, Chroma
from langchain_pinecone import PineconeVectorStore
```

## 9. RETRIEVERS

```
from langchain_core.vectorstores import VectorStoreRetriever
from langchain.retrievers import MultiQueryRetriever, ContextualCompressionRetriever
```

## 10. AGENTS

```
from langchain.agents import initialize_agent, AgentExecutor, create_openai_tools_agent
```

## 11. TOOLS

```
from langchain.tools import Tool
from langchain_core.tools import tool
```

## 12. OUTPUT PARSERS

```
from langchain_core.output_parsers import StrOutputParser, JsonOutputParser
```

## 13. MESSAGES

```
from langchain_core.messages import HumanMessage, AIMessage, SystemMessage
```

## MOST IMPORTANT

```
from langchain_core.prompts import ChatPromptTemplate
from langchain_openai import ChatOpenAI
from langchain_core.runnables import RunnableSequence
from langchain_core.output_parsers import StrOutputParser
```

## Modern Pipeline

```
prompt = ChatPromptTemplate.from_template("Explain {topic}")
model = ChatOpenAI()
parser = StrOutputParser()

chain = prompt | model | parser
```